

Java Programming Full Notes

Introduction to Java

Java is a high-level, object-oriented programming language developed by Sun Microsystems. Java is platform independent because programs run on JVM.

Features of Java

Java is Simple, Secure, Portable, Robust, Dynamic, Object-Oriented and Multithreaded.

JDK, JRE and JVM

JDK contains development tools, JRE provides runtime environment and JVM executes Java bytecode.

Variables and Data Types

Java supports int, float, double, char, boolean, long, short and byte data types.

Operators

Arithmetic, Relational, Logical, Assignment, Unary and Bitwise operators are used in Java.

Control Statements

if, if-else, switch, for loop, while loop and do-while loop are used for flow control.

Arrays

Arrays store multiple values of same type in contiguous memory locations.

Classes and Objects

A class is a blueprint while an object is an instance of the class.

Constructors

Constructors initialize objects and can be default or parameterized.

Inheritance

Inheritance allows one class to acquire properties and methods of another class.

Polymorphism

Method overloading and method overriding are examples of polymorphism.

Encapsulation

Encapsulation binds data and methods together in a single unit.

Abstraction

Abstraction hides implementation details and shows essential features only.

Interfaces

Interfaces provide complete abstraction and support multiple inheritance.

Exception Handling

try, catch, throw, throws and finally are used for exception handling.

Multithreading

Multithreading allows multiple tasks to execute simultaneously.

File Handling

Java supports file creation, reading and writing using File classes.

Collections Framework

Important collection classes are ArrayList, LinkedList, HashMap, HashSet and Stack.

JDBC

JDBC connects Java applications with databases using drivers.

Basic Java Syntax Examples

Concept	Example
Print Statement	<code>System.out.println("Hello World");</code>
Variable	<code>int a = 10;</code>
If Condition	<code>if(a > 5) { }</code>
For Loop	<code>for(int i=0;i<5;i++) { }</code>